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Grade

9

Integrated Science Revision Book

Book 1

First Edition 2026



Chemistry



Biology



Physics



Prepared by
Hezekiah Omweri Nyatuka
B.Ed. (Science), Kenyatta University
Chemistry & Physics

Peak Achievers Online Academy
Empowering Learners for a Brighter Future

PEAK ACHIEVERS ONLINE ACADEMY

GRADE 9

INTEGRATED SCIENCE

REVISION BOOK

*Comprehensive Notes • Practical Activities • Worked Examples • Topical Practice •
Assessment Preparation*

BOOK 1 • FIRST EDITION 2026

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Preface

This book is designed to help Grade 9 learners understand, revise and apply Integrated Science concepts. It brings together Chemistry, Biology and Physics using short, clear and content-rich notes, practical activities, worked scientific examples, topical practice and assessment preparation.

The emphasis is not only on remembering facts, but also on observation, investigation, interpretation, problem solving and evidence-based scientific reasoning.

How to Use This Book

- Read the revision notes carefully and study every labelled diagram.
- Work through the examples and practical activities.
- Attempt the checkpoint and topical questions before checking Part IV.
- Use the Mixed Revision Exercises to connect ideas from different chapters.
- Complete the five National Examination Practice Sets under timed conditions.
- Use the marking guidance to identify mistakes, then revise weaker areas.

EXAM FOCUS: In calculations, use the sequence: Formula → Substitution → Calculation → Answer + Unit.

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CHAPTER 1 — STRUCTURE OF ATOM

Learning Outcomes

- Describe the basic structure of an atom.
- Identify protons, neutrons and electrons and state their charges and locations.
- Use atomic number and mass number to determine numbers of subatomic particles.
- Write electron arrangements for the first 20 elements.
- Represent simple atoms using energy-level diagrams.

1.1 The Atom

An atom is the smallest particle of an element that retains the chemical identity of that element. Atoms contain a central nucleus surrounded by electrons arranged in energy levels.

Structure of an Atom

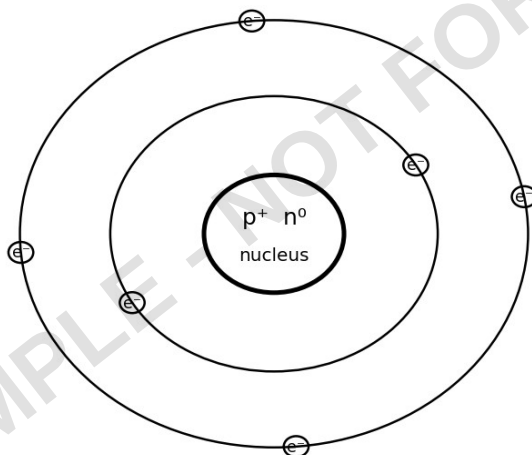


Figure: Structure of an atom

1.2 Subatomic Particles

Particle	Charge	Location
Proton	+1	Nucleus
Neutron	0	Nucleus
Electron	-1	Energy levels around nucleus

1.3 Atomic Number and Mass Number

Atomic number, Z , is the number of protons in the nucleus. Mass number, A , is the total number of protons and neutrons.

REMEMBER: $A = \text{protons} + \text{neutrons}$, therefore $\text{neutrons} = A - Z$. In a neutral atom, $\text{protons} = \text{electrons}$.

Worked Example

An atom has atomic number 15 and mass number 31.

Protons = 15

Neutrons = $A - Z = 31 - 15 = 16$

Electrons = 15

1.4 Electron Arrangement

For the first 20 elements, electrons are arranged in energy levels. Examples include 2,8,2 for atomic number 12; 2,8,5 for atomic number 15; 2,8,8 for atomic number 18; and 2,8,8,1 for atomic number 19.

Electron Energy-Level Diagrams

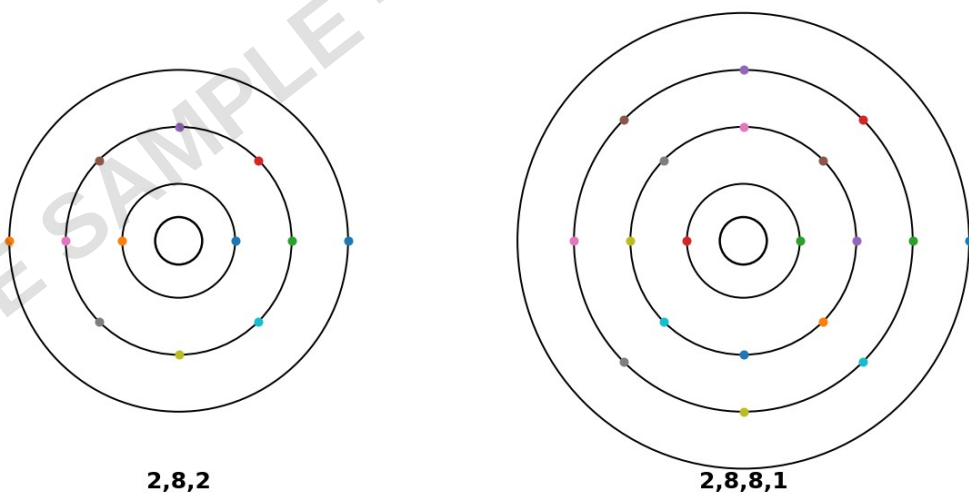


Figure: Electron energy-level diagrams

NB: At this level, focus on the structure of atoms and electron arrangements for the first 20 elements.

Checkpoint

1. State the charge and location of a neutron.
2. An atom has $Z = 14$ and $A = 28$. Determine its numbers of protons, neutrons and electrons.
3. Write the electron arrangement of an atom with atomic number 17.

Topical Practice — 20 Marks

1. Define an atom. (1 mark)
2. State the charge and location of each of the following: proton, neutron and electron. (3 marks)
3. Define atomic number. (1 mark)
4. Define mass number and write the relationship between mass number, protons and neutrons. (2 marks)
5. An atom has $A = 31$ and $Z = 15$. Calculate its number of neutrons. (2 marks)
6. State the relationship between the numbers of protons and electrons in a neutral atom. (1 mark)
7. Write the electron arrangements of atoms with atomic numbers 12, 15 and 18. (3 marks)
8. Draw and label an energy-level diagram for an atom with atomic number 19. (4 marks)
9. An atom contains 17 protons and 18 neutrons. Determine its atomic number and mass number. (2 marks)
10. State one reason why an atom is electrically neutral. (1 mark)

Leaf structure and stoma

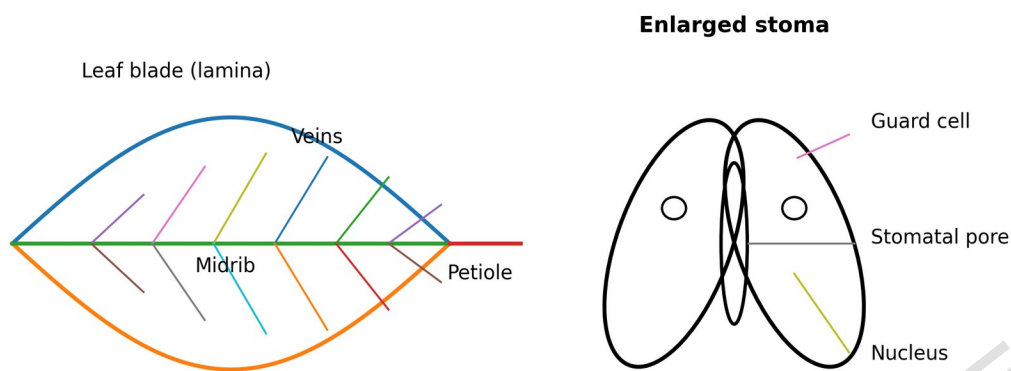


Figure: Leaf structure and stomata

4.3 Testing a Leaf for Starch

Aim: To test a leaf for the presence of starch.

Procedure: Place the leaf briefly in hot/boiling water. Decolourise it using ethanol in a water bath. Rinse/soften the leaf as appropriate, then add iodine solution.

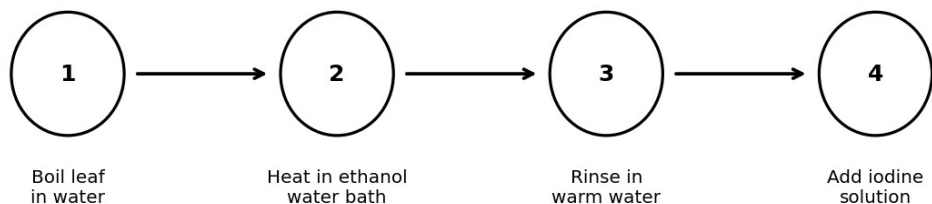
Positive result: Blue-black colour indicates starch.

Negative result: Brown or yellow-brown colour indicates that starch was not detected.

Safety: Ethanol is flammable. Do not heat it directly over a naked flame; use a water bath.

Starch Test for a Leaf

Positive: blue-black • Negative: brown/yellow-brown



SAFETY: ethanol is flammable — use a water bath; keep away from naked flame.

Figure: Starch test for a leaf

4.4 Investigating the Need for Light

A plant can first be destarched by keeping it in darkness for a suitable period. Part of a leaf is then covered so that light cannot reach it, while the remaining part is exposed to light. After exposure, the leaf is tested for starch.

Expected observation: the exposed region becomes blue-black while the covered region remains brown/yellow-brown.

CONCLUSION: Under the experimental conditions, light is necessary for photosynthesis/starch formation.

4.5 Investigating the Need for Chlorophyll

A variegated leaf contains both green and non-green regions. After suitable exposure to light, a starch test can compare these regions. Green regions are expected to show starch formation while non-green regions do not.

EXAM FOCUS: State what was observed before explaining what the observation means. Observation and interpretation are not the same thing.

CHAPTER 9 — WAVES

Learning Outcomes

- Explain a wave as a disturbance that transfers energy without permanent transfer of matter.
- Distinguish transverse and longitudinal waves.
- Identify amplitude, wavelength, crests, troughs, compressions and rarefactions.
- Use $v = f\lambda$, $T = 1/f$ and $f = 1/T$.
- Relate frequency to pitch and amplitude to loudness.
- Explain why sound requires a material medium.

9.1 Meaning of a Wave

A wave is a disturbance that transfers energy from one place to another without permanent transfer of matter.

9.2 Transverse and Longitudinal Waves

In a transverse wave, particles vibrate perpendicular to the direction of wave travel. In a longitudinal wave, particles vibrate parallel to the direction of wave travel.

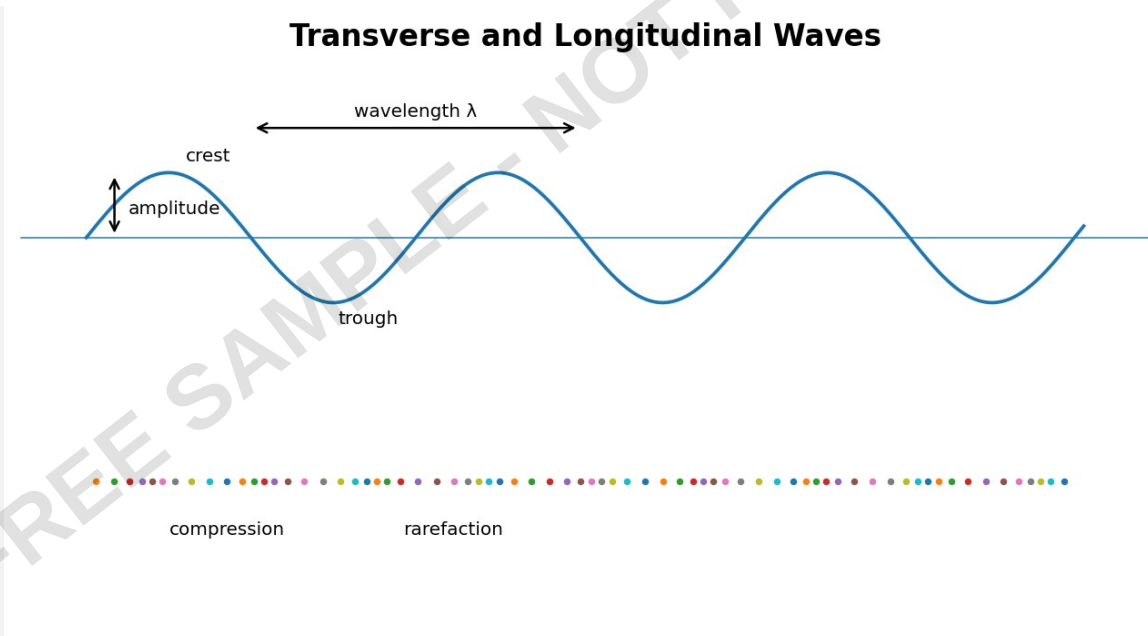


Figure: Transverse and longitudinal waves

9.3 Wave Terms

Amplitude: Maximum displacement from the equilibrium position.

Wavelength (λ): Distance between two successive corresponding points, such as crest to crest or compression to compression.

Frequency (f): Number of complete oscillations per second, measured in hertz (Hz).

Period (T): Time taken for one complete oscillation, measured in seconds.

REMEMBER: $T = 1/f$ and $f = 1/T$.

9.4 Wave Speed

$$v = f\lambda$$

Worked Example 1

For $f = 5$ Hz and $\lambda = 1.8$ m:

Formula: $v = f\lambda$

Substitution: $v = 5 \times 1.8$

Calculation: $v = 9$

Answer: $v = 9$ m/s

Worked Example 2

If 30 oscillations occur in 6 s:

$f = 30/6 = 5$ Hz

$T = 1/f = 1/5 = 0.2$ s

9.5 Sound Waves

Sound is produced by a vibrating source. In air, sound travels as a longitudinal wave involving compressions and rarefactions. Sound requires a material medium and cannot travel through a vacuum.

REMEMBER: Higher frequency generally corresponds to higher pitch. Greater amplitude generally corresponds to greater loudness.

9.6 Reflection of Sound

Sound can be reflected from suitable surfaces. A reflected sound may be heard separately as an echo when the conditions allow sufficient separation between the original and reflected sounds.

Practical Activity — Waves on a Rope

Aim: To observe a transverse wave and investigate frequency, wavelength and speed.

Procedure: Generate waves along a stretched rope under controlled conditions. Count oscillations over a measured time and measure wavelength.

Variables: If frequency is deliberately changed, it is the independent variable. Wavelength or another measured response may be the dependent variable. Keep the same rope and similar tension.

Reliability: Repeat timing and wavelength measurements; measuring several wavelengths together can reduce relative measurement error.

Conclusion: Use the data and $v = f\lambda$ to describe the relationship observed under the experimental conditions.

Data Example

Frequency (Hz)	Wavelength (m)	Speed (m/s)
2	4	8
4	2	8
8	1	8

Under these conditions, frequency increases while wavelength decreases, and the calculated wave speed remains 8 m/s.

Sound Data Example

Sound	Frequency	Relative amplitude
P	200 Hz	Small
Q	400 Hz	Small
R	400 Hz	Large

P has the lowest pitch. Q and R have the same pitch because they have the same frequency. R is louder because it has the greater amplitude.

Topical Practice — 20 Marks

1. Define a wave. (1 mark)
2. Distinguish transverse and longitudinal waves. (2 marks)
3. Name (a) the highest point of a transverse wave and (b) a region where particles are close together in a longitudinal wave. (2 marks)
4. Define wavelength. (1 mark)
5. A source makes 30 oscillations in 6 s. Calculate (a) frequency and (b) period. (4 marks)
6. Calculate the speed of a wave with frequency 5 Hz and wavelength 1.8 m. (2 marks)
7. State the relationship between (a) frequency and pitch and (b) amplitude and loudness. (2 marks)
8. Explain why sound cannot travel through a vacuum. (2 marks)
9. A wave travels at 12 m/s with frequency 8 Hz. Calculate its wavelength. (2 marks)
10. State two ways of improving reliability when measuring wave quantities. (2 marks)

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